Task 3: Model Validation, Overfitting Control & Hyperparameter Tuning

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1. Overfitting Analysis

An unconstrained Decision Tree Regressor was trained on the California Housing dataset (same features and preprocessing as Task 2) to illustrate overfitting. Without any depth limit, the tree grew until it perfectly memorized the training data, producing a train RMSE of essentially $0. However, its test RMSE was $69,790 — a massive gap that is a textbook sign of overfitting: the model learned the noise in the training data rather than generalizable patterns, so it failed to perform well on unseen data. This confirms that high training accuracy alone is meaningless if the model cannot generalize.

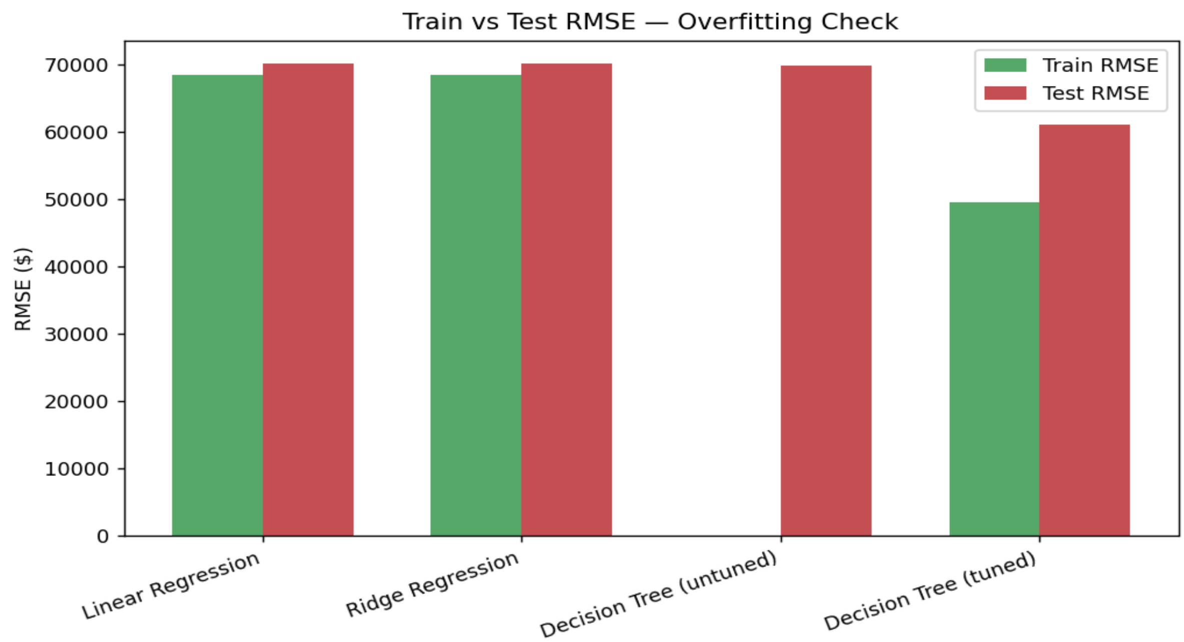

Figure 1: Train vs Test RMSE across all models. The untuned Decision Tree shows a huge gap (severe overfitting); Linear/Ridge Regression show almost no gap (they are simple and don't overfit, but also underfit the non-linear signal).

2. Cross-Validation & Hyperparameter Tuning Approach

To get a more reliable performance estimate than a single train/test split, 5-fold cross-validation was applied. The untuned tree's cross-validated RMSE was $99,585 (± $11,320), confirming its instability. GridSearchCV was then used to systematically search over max_depth ([3, 5, 7, 10]) and min_samples_split ([2, 5, 10]) — 12 combinations, each evaluated with 5-fold cross-validation — to find the hyperparameters that minimize RMSE without letting the tree overfit. The best combination found was max_depth=10, min_samples_split=10.

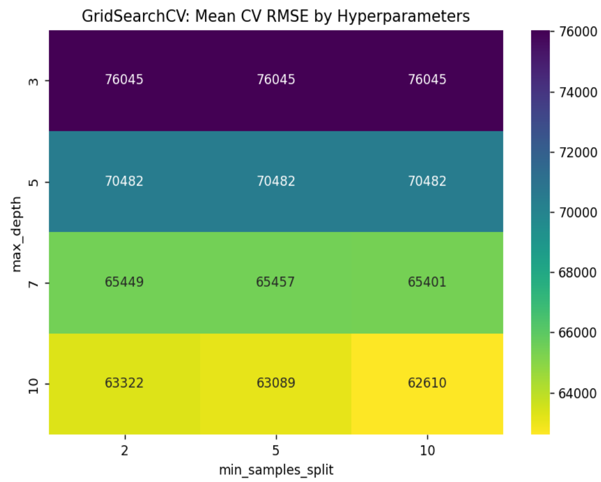

Figure 2: GridSearchCV results — mean cross-validated RMSE for every hyperparameter combination tested. Darker cells are better (lower RMSE).

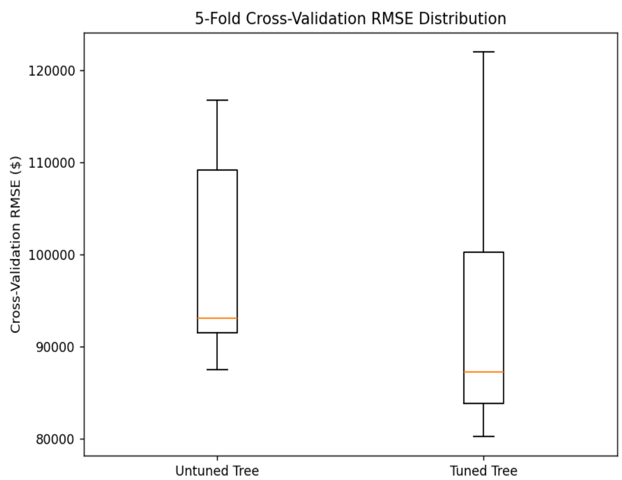

Figure 3: Cross-validation RMSE distribution before and after tuning. The tuned model's scores are lower.

3. Results & Model Comparison

Model	Train RMSE	Test RMSE	Test R^2	Overfit Gap
Linear Regression	68,435	70,061	0.6254	1,627
Ridge Regression	68,434	70,057	0.6255	1,623
Decision Tree (untuned)	~0	69,790	0.629	69,790
Decision Tree (tuned)	49,498	60,997	0.7161	11,499

(Tuned Decision Tree row highlighted — best test RMSE and R^2, with a far smaller overfitting gap than the untuned tree.)

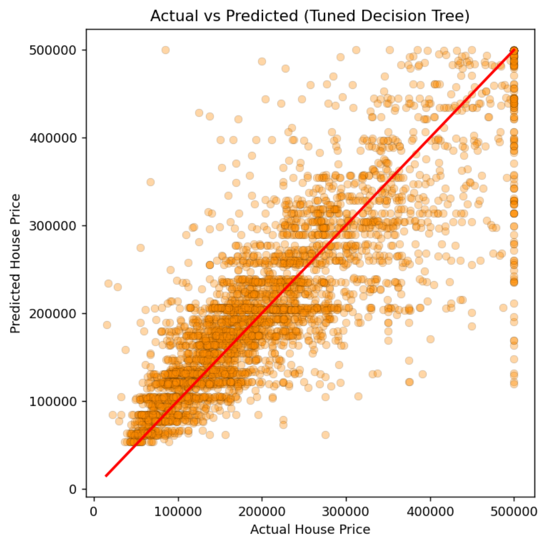

Figure 4: Actual vs Predicted values for the final tuned model. Points cluster closer to the red reference line than the Task-2 linear models did.

4. Final Model Selection Justification

Why this model was selected: The tuned Decision Tree (max_depth=10, min_samples_split=10) achieved the lowest test RMSE (~$60,997) and highest R^2 (~0.716), outperforming both linear baselines and the untuned tree.
How overfitting was reduced: Constraining max_depth and min_samples_split via GridSearchCV shrank the overfitting gap from ~$69,790 (untuned) to ~$11,499 (tuned), while simultaneously improving test performance.
Why cross-validation results are trusted: Evaluating across 5 different train/test splits, rather than one, confirms the reported performance is consistent and not a result of a lucky split — standard practice before deploying a model.
Trade-offs: Linear/Ridge Regression are simpler and fully interpretable but cannot capture non-linear relationships, capping their accuracy. The tuned Decision Tree is less interpretable and costs more to tune, but generalizes well and delivers meaningfully higher accuracy — the right trade-off here since predictive accuracy is the priority.

5. Deliverables

AI_ML_Task3_Model_Validation_Tuning.ipynb (cross-validation, GridSearchCV tuning, overfitting analysis, comparison table, plots), this PDF report, and the saved tuned model (best_tuned_model.joblib + scaler.joblib).